FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

IT0049
(WEB SYSTEM TECHNOLOGIES)

TECHNICAL FORMATIVE ASSESSMENT

3

MAKING IT EDITABLE: FORMS, VALIDATION, AND FILE UPLOAD

Student Name / Group Name:	Vincent Luke Elpedez
Members (if Group):	**Name** / **Role**
Section:	TC36

Professor:	Sir, Magbitang

PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements. [PO: C]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Apply advanced web development principles in creating, debugging, validating, and securing database-backed web applications. [CLO: 2]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Build and validate HTML forms for creating new records.
- Redisplay a form with validation errors and the user's previously entered values.
- Implement an edit and update workflow for existing records.
- Accept, validate, prepare, and store an uploaded image file, saving only its filename in the database.

BACKGROUND INFORMATION

A read-only application is only half useful. This activity adds the ability to create and edit records through validated forms — rejecting bad input instead of trusting it — and introduces file upload, a common requirement for anything involving user avatars or product images. CodeIgniter's validation rules, getFile() method, and Image service handle these concerns without changing the main upload workflow.

GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (40 points max)	All required pages and features are present and work as specified in the activity instructions.
Code Structure & Organization (25 points max)	Controllers, models, and views are separated correctly and follow the framework's conventions; code is readable and sensibly named.
Form Validation & File Handling (20 points max)	Forms reject invalid input with clear error messages and preserve the user's entries on redisplay; the uploaded file is validated by type and size, prepared for display, stored safely, and only its filename is saved to the database.

CRITERIA	Description
Documentation & Submission Quality (15 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.

LABORATORY ACTIVITY

Extend the POS application from TFA2 with the ability to create and edit customer and user accounts, and add avatar upload with display-ready image preparation for user accounts.

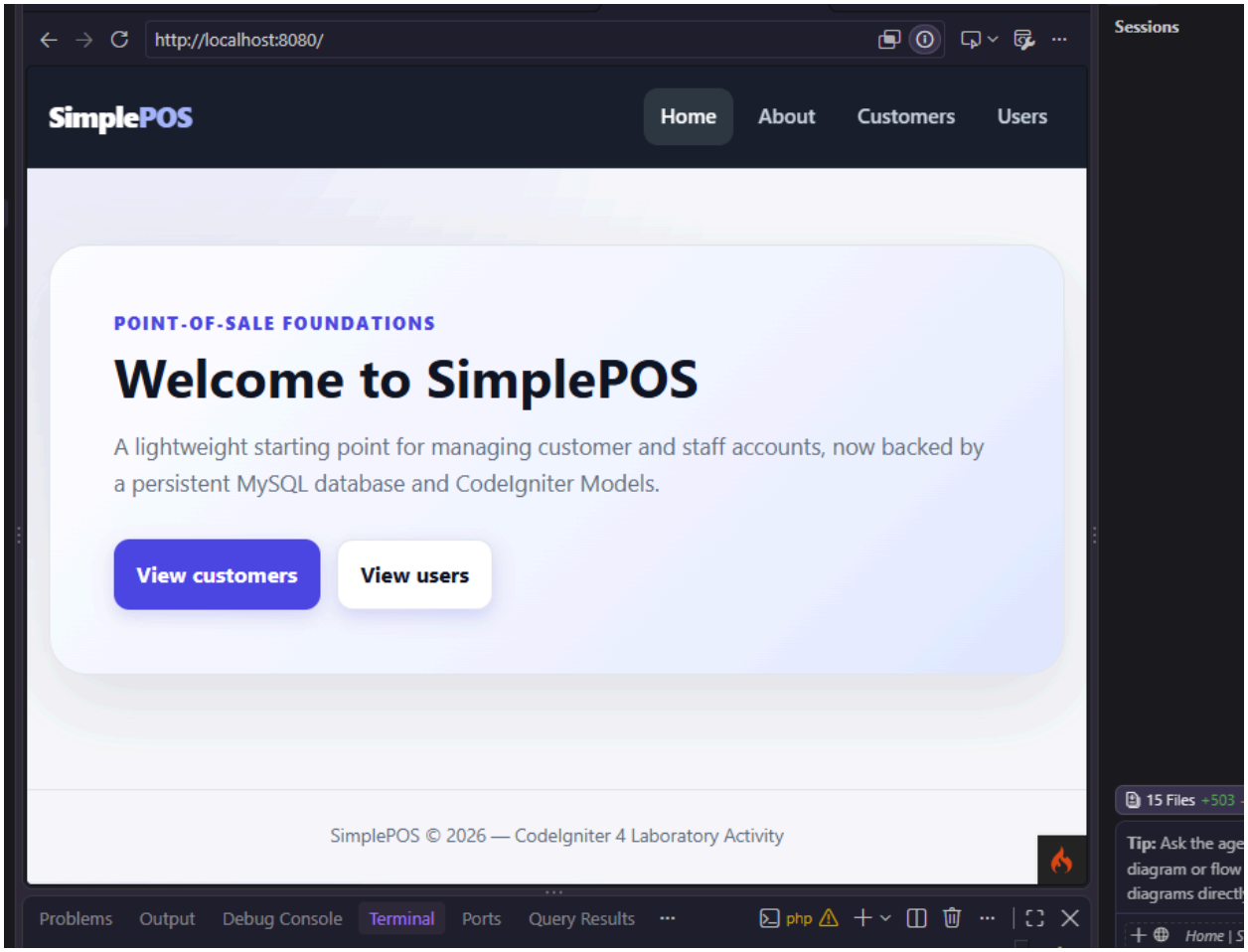
1. Add a New Customer form (/customers/new) that validates required fields (full name required, email required and valid) before inserting a record.
2. Add a New User form (/users/new) that validates required fields (username required and unique, full name required) before inserting a record.
3. Add edit pages for both customer and user accounts that pre-fill the form with the existing record's data and update it on submit.
4. Add an avatar column to the users table. On the user edit form, add a file input for a profile picture; validate that the uploaded file is a JPG or PNG no larger than 2MB, prepare a display-ready copy or thumbnail, store it in a public uploads folder, and save only the filename in the avatar column.
5. Display each user's prepared avatar or thumbnail on the User Accounts listing page (fall back to a placeholder image if none was uploaded).

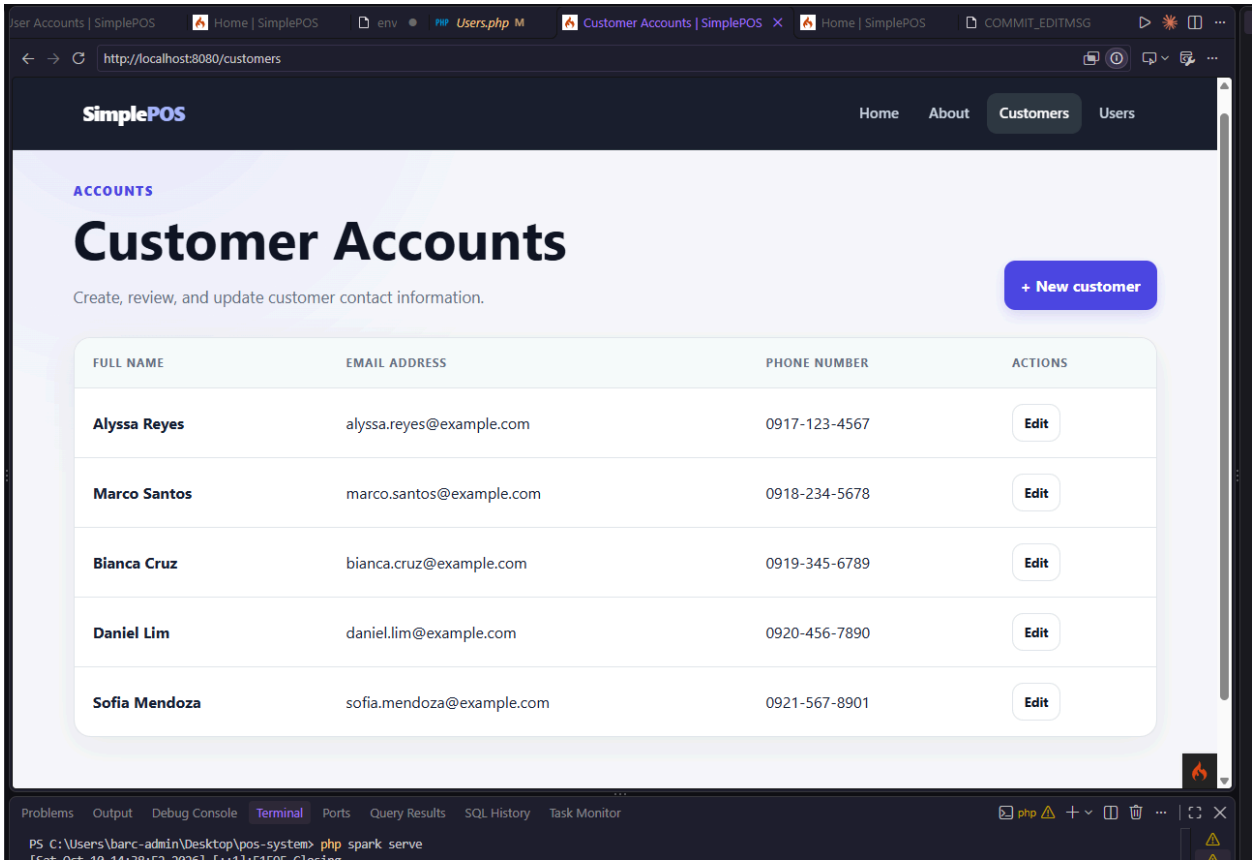
Submission Requirements:

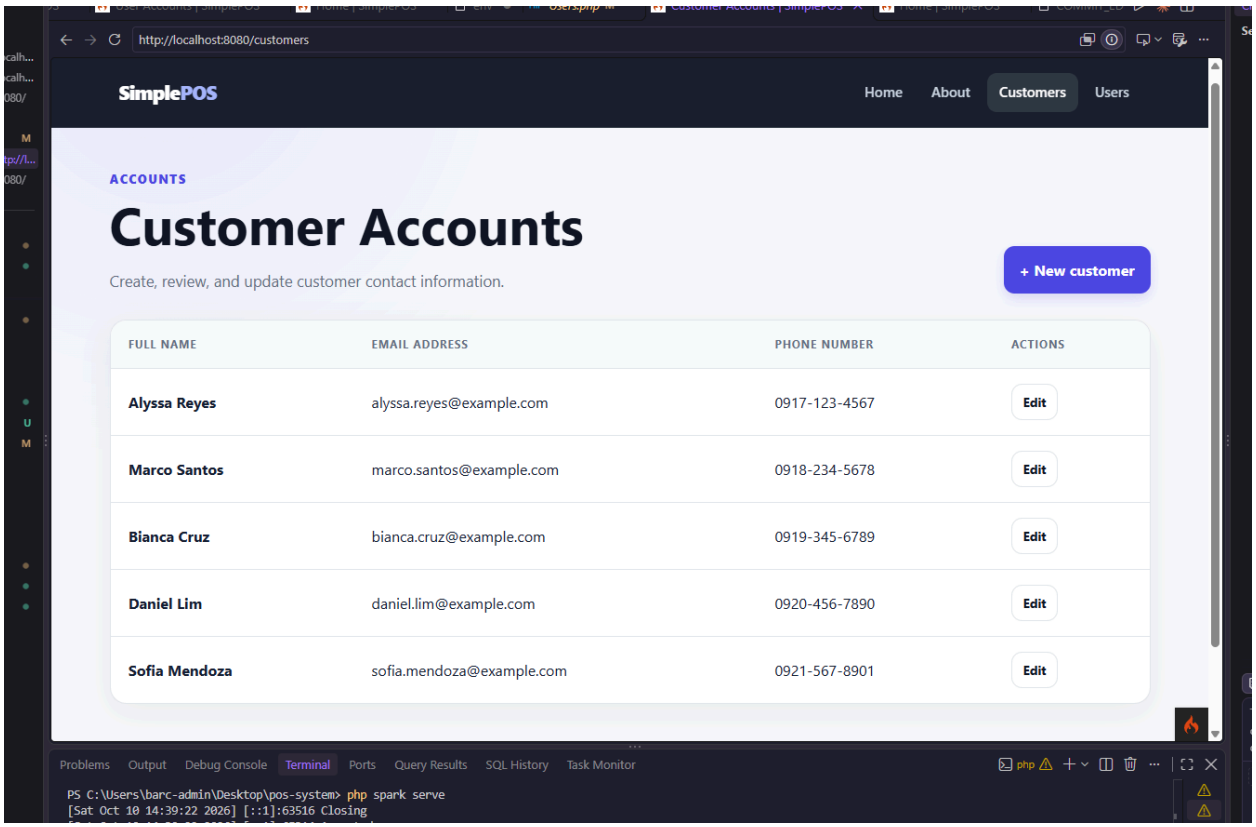
- A link to a GitHub repository containing your raw project files, including your database export.
- A link to your hosted, working version of the application.

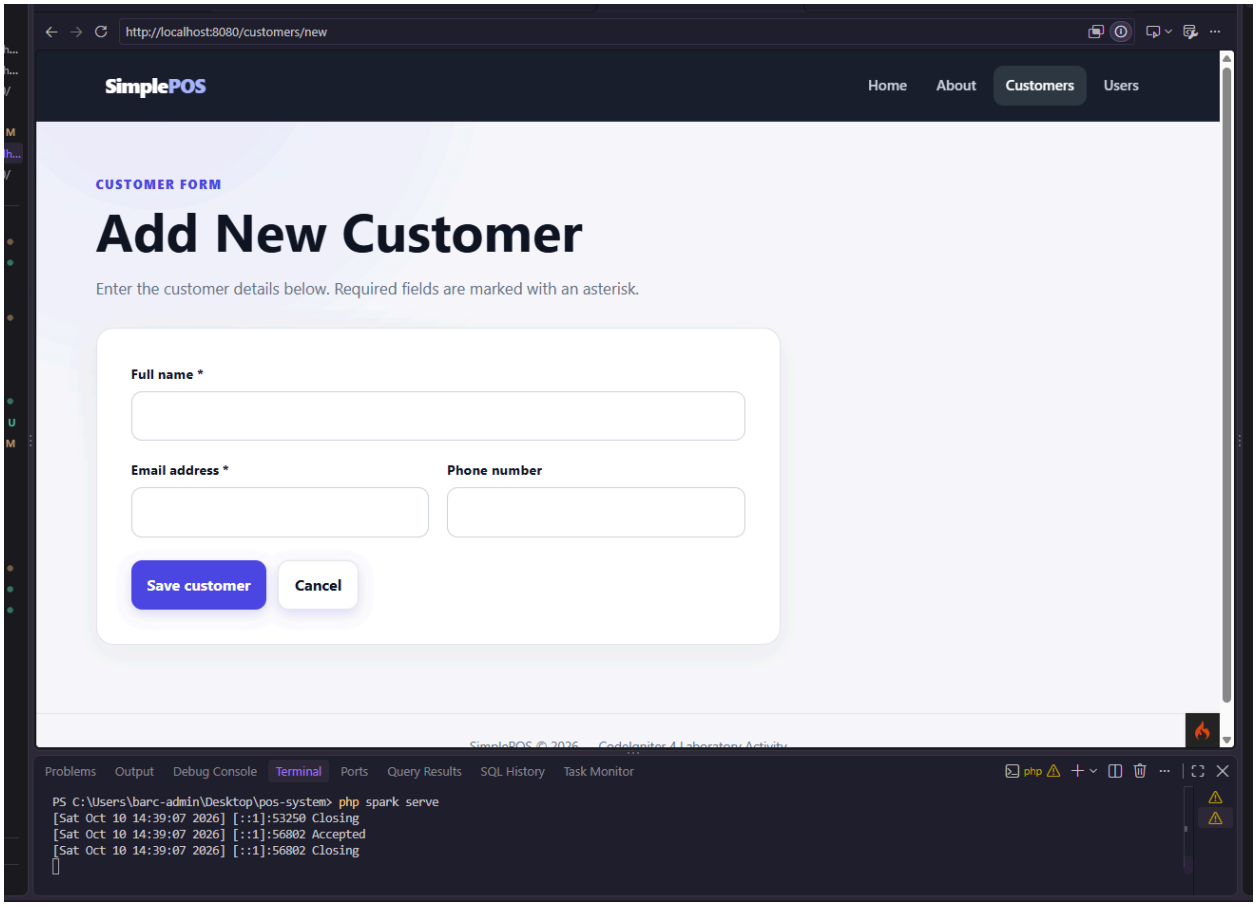
REFERENCES

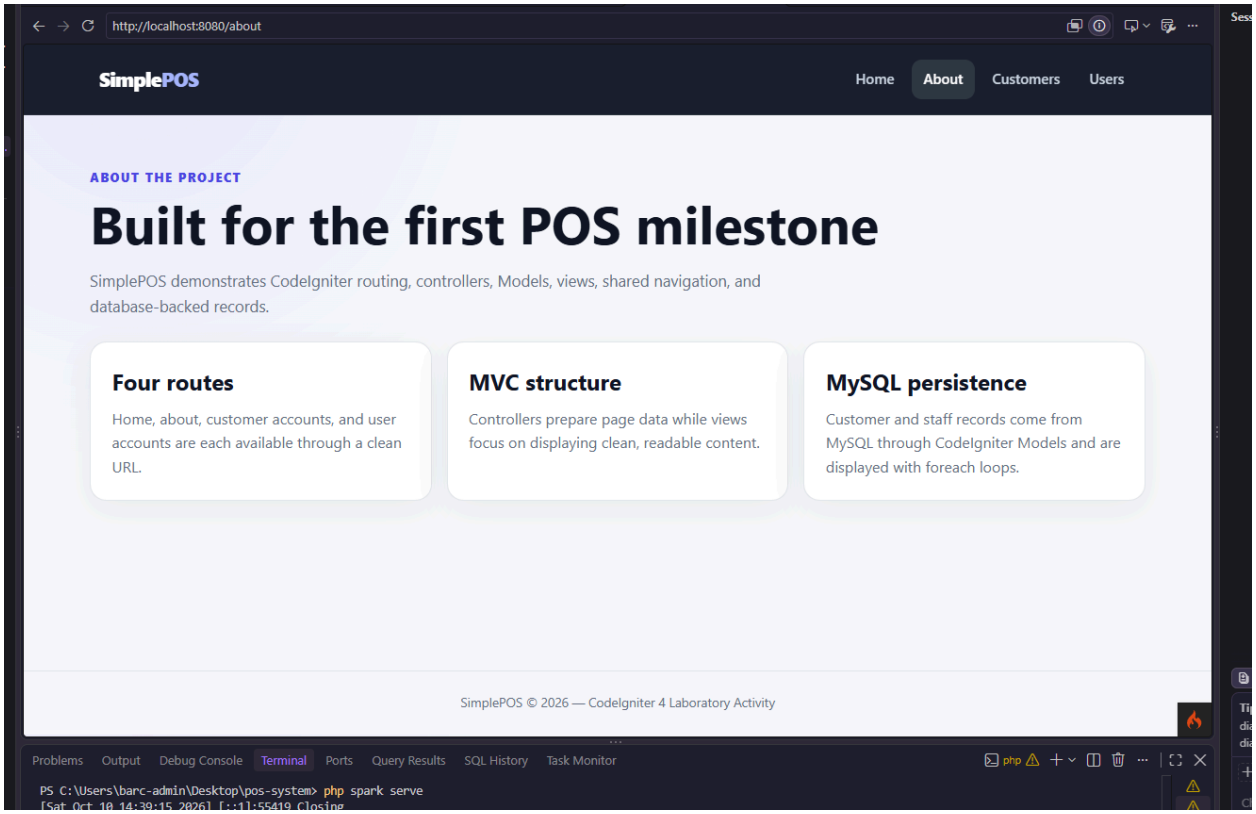
- CodeIgniter User Guide. Validation. https://codeigniter4.github.io/userguide/libraries/validation.html
- CodeIgniter User Guide. File Uploads. https://codeigniter4.github.io/userguide/libraries/uploaded_files.html

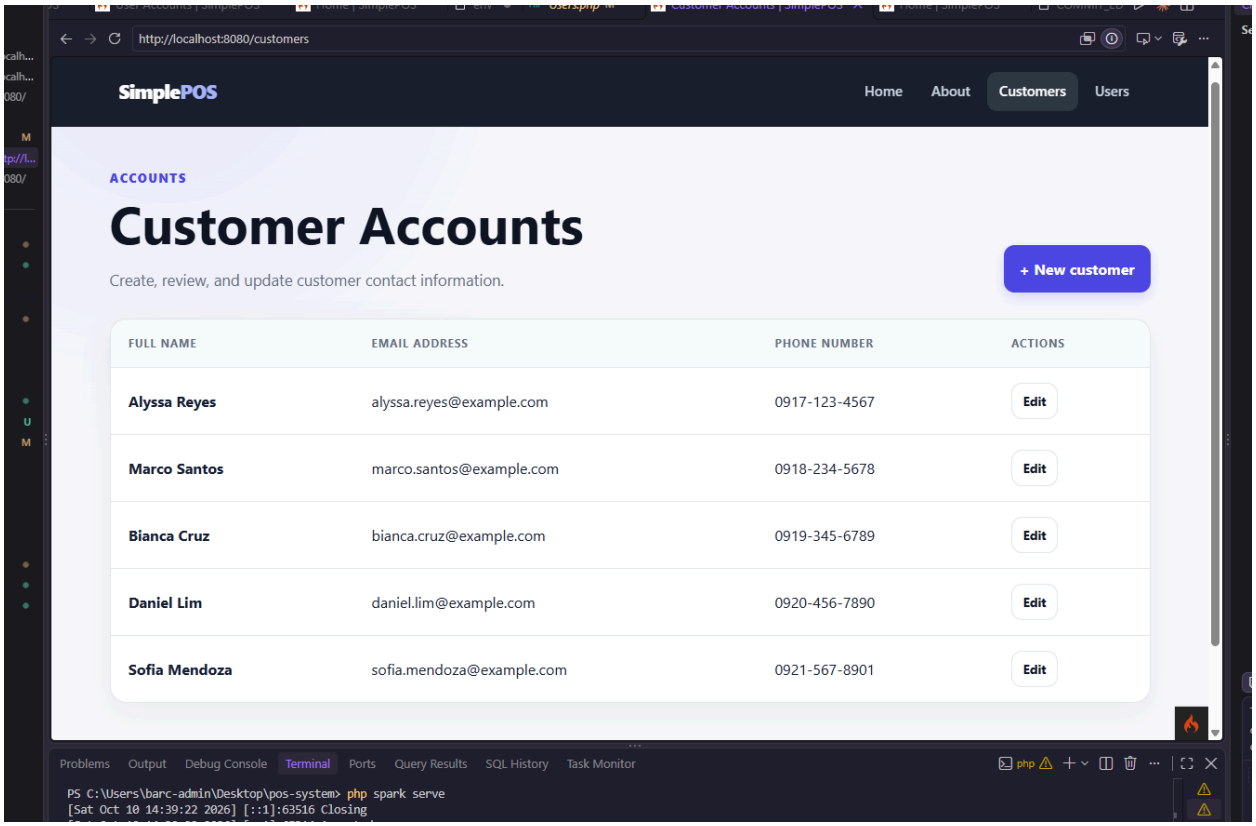

http://localhost:8080/
SimplePOS
Home
About
Customers
Users
POINT-OF-SALE FOUNDATIONS
Welcome to SimplePOS
A lightweight starting point for managing customer and staff accounts, now backed by a persistent MySQL database and CodeIgniter Models.
View customers
View users
SimplePOS © 2026 — CodeIgniter 4 Laboratory Activity
Problems
Output
Debug Console
Terminal
Ports
Query Results
php
Sessions
15 Files +503
Tip: Ask the age
diagram or flow
diagrams directl


http://localhost:8080/customers
SimplePOS
Home
About
Customers
Users
ACCOUNTS
Customer Accounts
Create, review, and update customer contact information.
+ New customer
FULL NAME
EMAIL ADDRESS
PHONE NUMBER
ACTIONS
Alyssa Reyes
alyssa.reyes@example.com
0917-123-4567
Edit
Marco Santos
marco.santos@example.com
0918-234-5678
Edit
Bianca Cruz
bianca.cruz@example.com
0919-345-6789
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Daniel Lim
daniel.lim@example.com
0920-456-7890
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Sofia Mendoza
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0921-567-8901
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PS C:\Users\barc-admin\Desktop\pos-system> php spark serve


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[Sat Oct 10 14:39:22 2026] [::1]:63516 Closing


http://localhost:8080/customers/new
SimplePOS
Home
About
Customers
Users
CUSTOMER FORM
Add New Customer
Enter the customer details below. Required fields are marked with an asterisk.
Full name *
Email address *
Phone number
Save customer
Cancel
Problems
Output
Debug Console
Terminal
Ports
Query Results
SQL History
Task Monitor
PS C:\Users\barc-admin\Desktop\pos-system> php spark serve
[Sat Oct 10 14:39:07 2026] [::1]:53250 Closing
[Sat Oct 10 14:39:07 2026] [::1]:56802 Accepted
[Sat Oct 10 14:39:07 2026] [::1]:56802 Closing


http://localhost:8080/about
SimplePOS
Home
About
Customers
Users
ABOUT THE PROJECT
Built for the first POS milestone
SimplePOS demonstrates CodeIgniter routing, controllers, Models, views, shared navigation, and database-backed records.
Four routes
Home, about, customer accounts, and user accounts are each available through a clean URL.
MVC structure
Controllers prepare page data while views focus on displaying clean, readable content.
MySQL persistence
Customer and staff records come from MySQL through CodeIgniter Models and are displayed with foreach loops.
SimplePOS © 2026 — CodeIgniter 4 Laboratory Activity
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